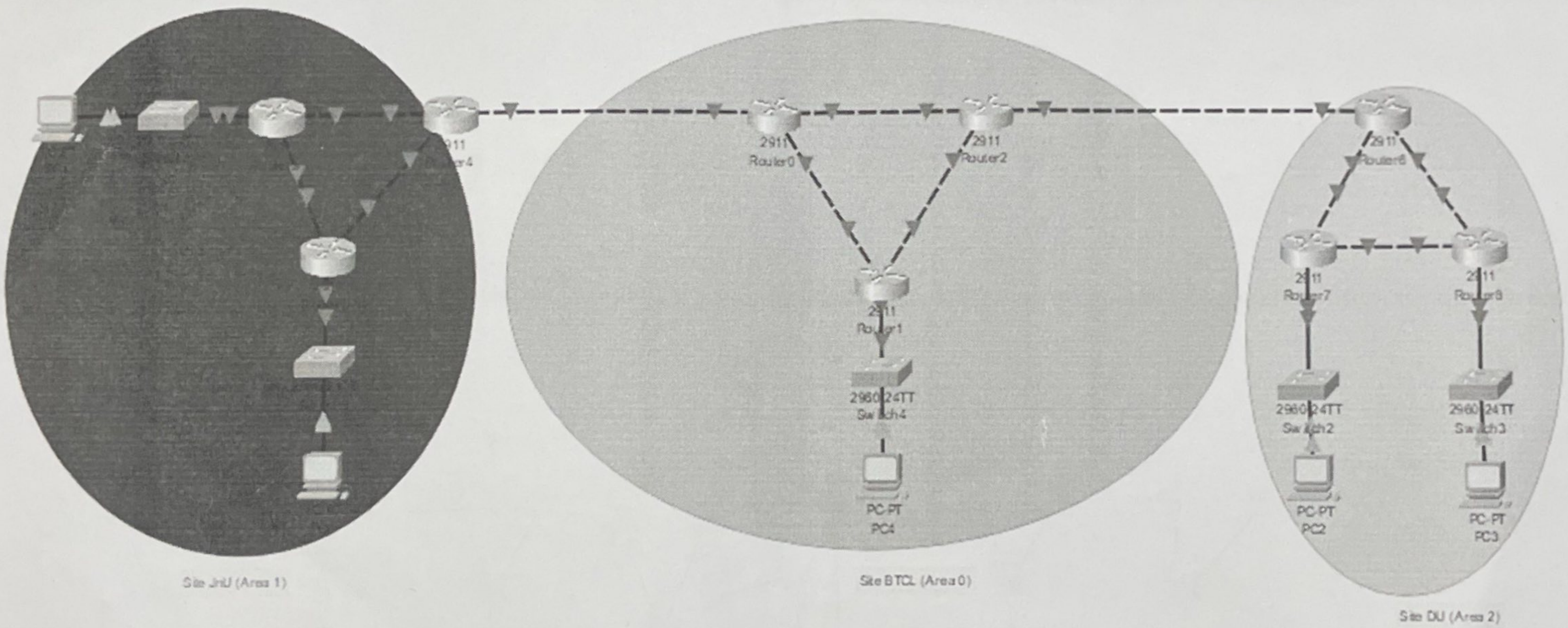
Set 02

Kishoreganj University
3rd Year 2nd Semester B.Sc. (Engg.) Lab Final Examination-2024
Department of Computer Science and Engineering
CSE3202: Computer Networking Lab

Time: 1 Hour **Answer All the Questions** **Marks: 40**

1.



OSPF

Consider the above network topology containing three sites (JnU, BTCL, and DU). The allocated blocks of three sites are as follows:

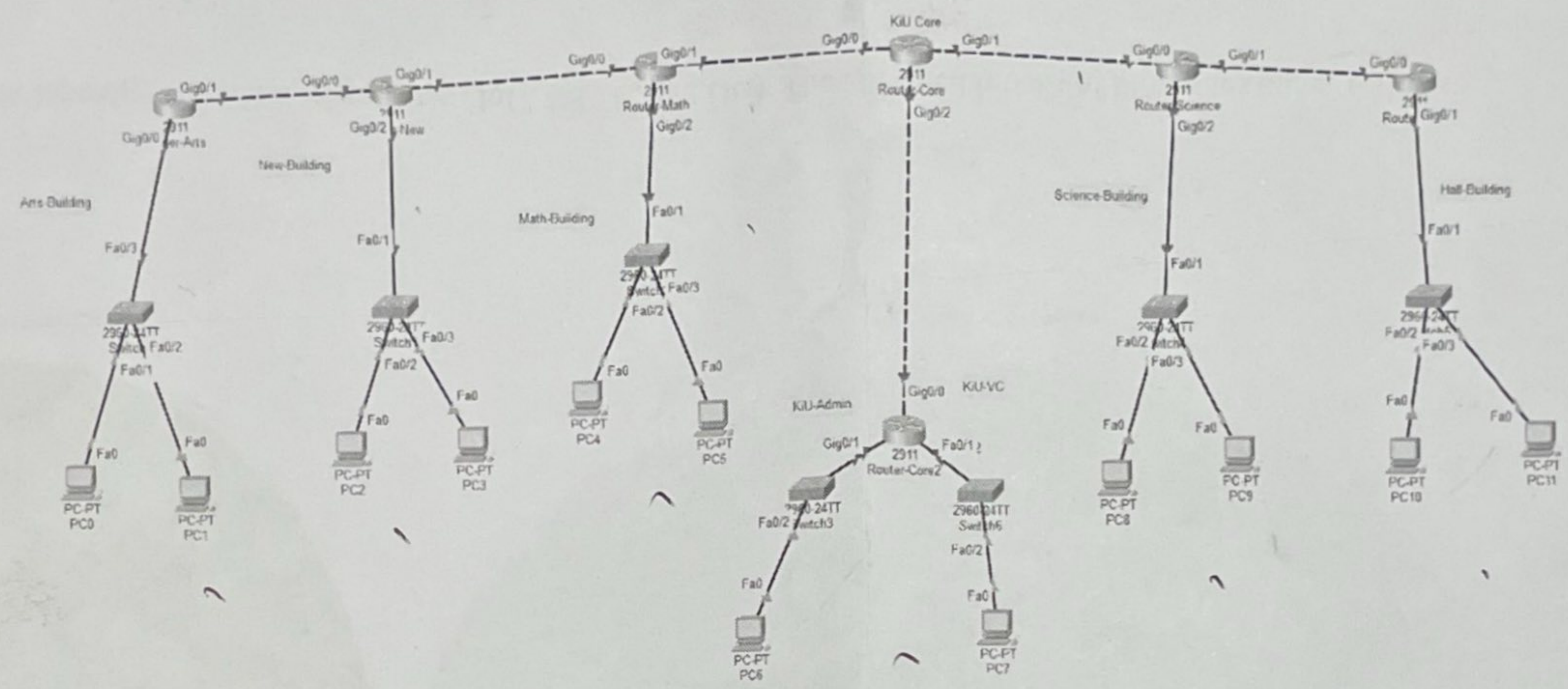
JnU: 1XY.XY.0.0/20

BTCL: 1YX.YX.0.0/16

DU: 1XY.YX.0.0/20

Design the subblocks and give the slash notation for each subblock in different sites. All the point-to-point connections address should be allocated from this granted block. Implement the network topology in packet tracer and test data transmission among different networks using MOSPF routing protocol.

2.



BTCL has granted Jagannath University a block of addresses starting with **1YX.XY.0.0/20**. Jagannath University wants to distribute these blocks to different academic and administrative buildings as follows:

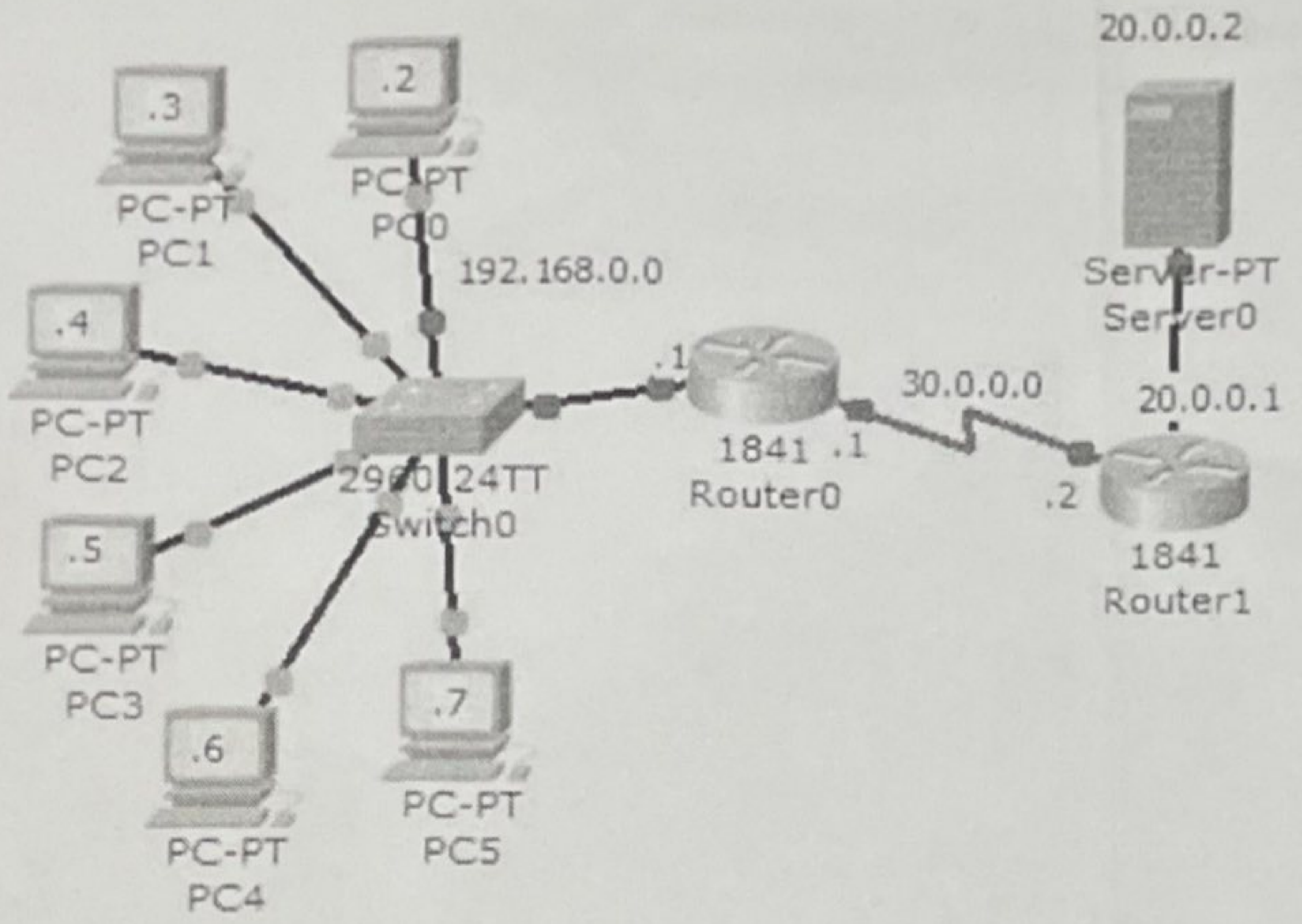
1. Arts Building: 300 addresses
2. New Building: 500 addresses
3. Math Building: 200 addresses
4. Admin Building: 250 addresses
5. VC Building: 350 addresses
6. Science Building: 600 addresses
7. Hall Building: 200 addresses

Design the subblocks and give the slash notation for each subblock. All the point-to-point connections address should be allocated from this granted block. Find out how many addresses are still available after these allocations.

Implement the network topology in packet tracer and test data transmission among different networks using **RIPv2 routing protocol**.

Site Address: 1YX.XY.0.0/20 [XY represents last two digits of your student ID]

3.

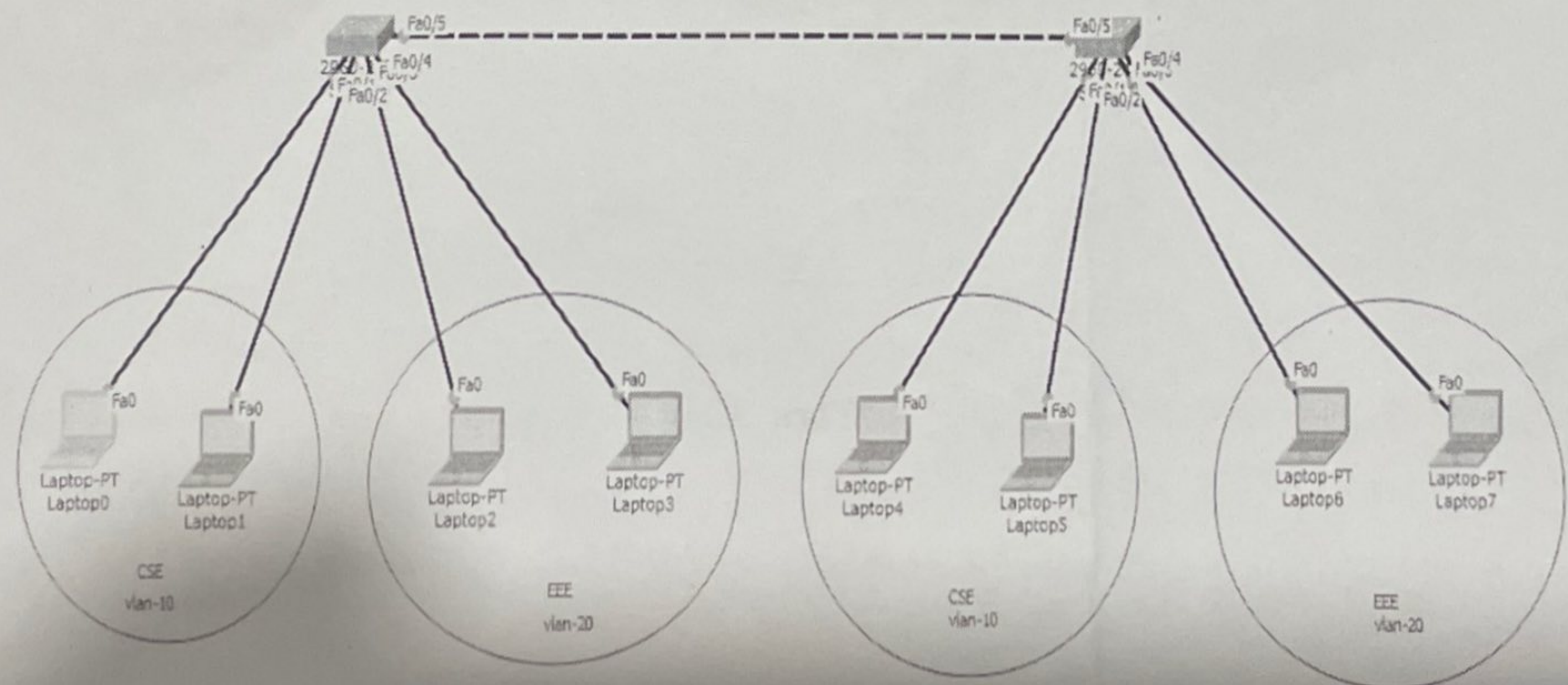


Implement the network topology in Packet Tracer and nsure that all internal devices can communicate with the external network using NAT.

4.

5

192.168.1.1-10 CSE
192.168.1.11-20 EEE



Test connectivity within the same VLAN and across different VLANs using ping.